Theory of computation (MC 223)

Course placement: It is a core course for the BTech MnC

Course format: It is 3 hours lecture every week and 1 hour tutorial.

Course content: Mathematical notations, different types of proofs, regular languages, DFA, NFA, regular expressions, context-free grammar, pushdown automaton, Turing machines and un-decidability.

Text book: Introduction to Theory of Computation, 2 nd Ed, Michael Sipser, Wadsworth Publishing Co Inc, 2012.

Reference books: Languages and Machines, 3 rd Ed, Thomas A. Sudkamap, Addison Wesley, 2006.

An introduction to Formal Languages and Automaton, Narosa Publishing House, 2007

Assessment method: A student will be awarded based on the marks obtained by him in the mid semester and the end semester.

Course outcomes: This subject forms the theoretical foundation of computer science. While the contemporary languages and architectures would become obsolete someday, the foundational matter in computer science would always remain relevant. Studying this subject will realize the student about the capabilities and limitations of computer. It also tells the students about the kind of problems that are realistically tractable.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X								X

Lecture schedule:

Mathematical notations: 2 lecture

Proof types: 2 lectures

Deterministic FSA: 4 lectures

Non-deterministic FSA: 4 lectures

Regular expressions: 2 lecture

Non regular languages: 3 lectures

Properties of regular languages: 3 lectures

Context-free grammar: 4 lectures

Pushdown automata: 3 lectures

Non context-free languages: 3 lectures

Turing machines: 3 lectures

Decidability: 3 lectures

Undecidability: 4 lectures

Grading policy (tentative): The grade earned by a student will depend on (1) the total marks earned by him in all the exams conducted over the entire semester, and (2) active participation in the class discussions. The weightage of each part is as follows:

Mid sem marks: 40

End sem marks: 40

Class participation: 20 %